Customer Shopping Behaviour Analysis

1. Project Overview

This project explores customer purchasing patterns using a dataset of 3,900 shopping transactions across multiple product categories. The analysis focuses on understanding customer preferences, spending habits, repeat purchases, and subscription trends to help business make informed decisions.

2. Dataset Information

The dataset consists of 3,900 records and 18 columns.

Main attributes include:

- Customer Details: Age, Gender, Location, and Subscription Status
- Transaction Details: Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping patterns: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

The dataset initially contained 37 missing entries in the review rating column.

3. Exploratory data analysis using python

Python was used to prepare the dataset and ensure it was suitable for further analysis.

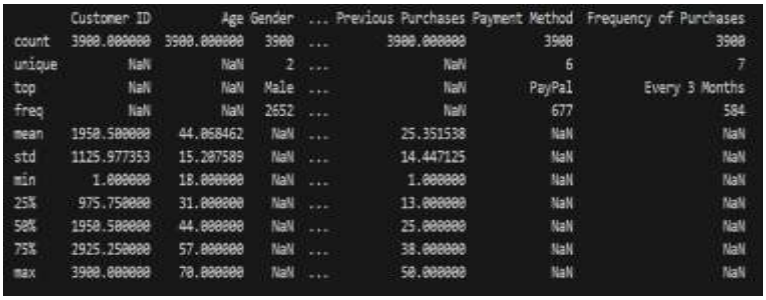
The following activities were performed:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.



#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	str
3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)



	Customer ID	Age	Gender	...	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	...	3900.000000	3900	3900
unique	NaN	NaN	2	...	NaN	6	7
top	NaN	NaN	Male	...	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	...	NaN	677	584
mean	1950.500000	44.068462	NaN	...	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	...	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	...	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	...	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	...	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	...	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	...	50.000000	NaN	NaN

- **Missing Data Handling:** Identified missing values and filled missing review ratings using the median rating of the corresponding product category.
- **Column standardization:** Renamed columns using snake case for consistent naming.
- **Feature Engineering:**
  - Created age_group column for demographic analysis.
  - Created Purchase_frequency _days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4.Data Analysis using SQL (Business Transactions)

MySQL queries were used to investigate customer behaviour and answer important business questions.

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

gender	revenue
Male	157890
Female	75191

2. **High-spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

10. . **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5.Dashboard in Power BI

Power BI was used to develop an interactive dashboard that presents the analysis visually.



6. Business Recommandations

- **Increase Subscription Adoption:** Communicate the benits of subscribing to encourage more customers to join.
- **Improve Customer Retention:** Introduce rewards and incentives for frequent shoppers.
- **Optimize Discount Stragies**: Evaluate discount effectiness while considerirng profitability.
- **Promote Popular Products**: Feature highly rated and frequently purchased products in marketing compaigns.
- **Personalize Marketing:** Tailor promotional activities to relevant age groups and customer segments.